

Group 1 :

Write the scientific term expressed by the following statements:

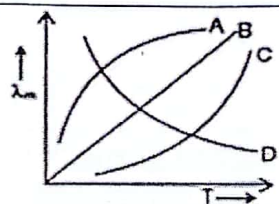
	Statement	
1	The total number of magnetic flux lines passing through a surface	Magnetic flux
2	The point at which the total magnetic flux density vanishes	Neutral point
3	The magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field	Tesla
4	The induced current must be in a direction such as to oppose the change producing it	Lenz's law
5	They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect)	Eddy currents
6	The distribution of the radiation intensity with wavelength is	Planck's distribution
7	The wave length at maximum radiation intensity ($\lambda_{\max}$) is inversely proportional to the temperature	Wien's law
8	emission of electrons from a heated metal's surface (by heating)	Thermionic emission
9	The ratio of output electric power to the input electric power.	Efficiency of a transformer
10	It's the electromagnetic effect takes place between two coils when an induced emf generated in one of them due current variation in the other coil	Mutual induction between two coils

باقي الإجابة في الورقة التالية

Group 2 :**Put circle around the correct answer**

11	A rectangular wire loop of its dimensions (5cm × 8cm) is at rest in a uniform magnetic field of flux density 2T that is directed into the page. The plane of the loop is perpendicular to the field. The total magnetic flux through the loop is..... a) Zero b) 0.002 Wb c) 0.0052 Wb d) 0.008 Wb
12	If the magnitude of the magnetic field a distance d from a very long, straight, current-carrying wire is B, at what distance from the wire will the field have magnitude 4B? a) 8d b) 4d c) d/4 d) d/8
13	The magnetic flux density increases at the center of a circular coil carrying an electric current by decreasing a) cross sectional area of the coil. b) the number of the coil turns. c) the current intensity in the coil d) the permeability of the coil core.
14	A solenoid is carrying electric current. As an iron core is inserted into the solenoid, the magnetic field density of the electromagnet formed a) decreases b) increases c) remains the same d) zero
15	The force on a wire carrying current placed in a uniform magnetic field is the maximum when..... a) The current is parallel to the field lines. b) The current is at a 60° angle with respect to the field lines. c) The current is at a 30° angle with respect to the field lines. d) The current is perpendicular to the field lines
16	A current-carrying, rectangular coil of wire is placed in a magnetic field. The magnitude of the torque on the coil <u>is not dependent upon</u> which one of the following quantities? a) The magnitude of the current in the loop b) The area of the loop c) The direction of the current in the loop d) The orientation of the loop
17	Which of the following will cause an induced current in a coil of wire? a) A magnet resting near the coil. b) The constant field of the earth passing through the coil. c) A magnet being moved into or out of the coil. d) A wire carrying a constant current near the coil
18	A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is parallel to the magnetic field is (V). If the frequency of rotation is halved, the value of instantaneous value of emf produced at the same position is..... a) ¼V b) ½V c) V d) 2V

باقي الإجابة في الورقة التالية

19	For a black body radiation if $\lambda_{\max}$ represents the maximum wavelength corresponding to the maximum radiation at temperature T, the curve between $\lambda_{\max}$ and T is	
	a) A b) B c) C d) D	
20	If a transformer is assumed to be an ideal one, all the following statements are correct EXCEPT	
	a) The current ratio of secondary to primary is equal to the turn's ratio of secondary to primary.	
	b) The current ratio of secondary to primary is equal to the turn's ratio of primary to secondary.	
	c) The voltage ratio of secondary to primary is always equal to the turn's ratio of secondary to primary.	
	d) The electric power produced at the secondary coil is equal to the electric power given in the primary coil	

Group 3 :**Mention the rule or the method used to determine**

	Statement	
21	The direction of the magnetic field resulting from an electric current in a wire	Ampere Right Hand Rule
22	The direction of the magnetic field resulting from an electric current in a circular coil	The right Cork Screw rule (Maxwell's Screw Rule)
23	The direction of the force (motion) of a straight wire carrying current placed in a uniform magnetic field	Fleming left hand rule
24	The direction of induced current in moving wire in a uniform magnetic field	Fleming Right hand rule
25	image moving objects in the dark due to the heat radiation which these objects re - emit	(Thermal imaging) (remote Sensing) night vision system
26	Production of backward induced electromotive force (emf) on secondary coil when the circuit of primary coil is closed	a) Increasing the current intensity in primary coil by using rheostat. b) moving the primary coil near to or inside secondary coil

باقي الإجابة في الورقة التالية

Group 4 :**Mention one factor that can increase**

	Statement	
27	The density of magnetic field at a certain point from a straight wire carrying current.	a) increase intensity of current passing through the wire b) decrease the distance of the point from the wire
28	The magnetic force acting on a wire carrying current placed normally in a uniform magnetic field.	a) increase the length of the wire b) increase the current in the wire c) increase the magnetic flux density
29	The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field	a) increase magnetic flux density b) increase the current intensity c) increase the area of the coil's plane d) increase number of turns of the coil
30	The induced emf in a straight wire moving normal to a magnetic field	a) increase the wire of length b) increase the magnetic field of density c) increase velocity of motion
31	The induced voltage produced from the electric transformer	a) By Using step up transformer b) By making the transformer ratio more than one
32	The velocity of electron emitted from the cathode in cathode ray tube	increasing the potential difference between the cathode and anode

باقي الإجابة في الورقة التالية

Group 5:

Answer the following

	problem	
33 34	In an ideal transformer, the secondary coil has twice as many turns as the primary. If the alternating current in the primary coil is 2 A. calculate the current in the secondary coil is	$\frac{N_s}{N_p} = \frac{I_p}{I_s}$ $\frac{2N_p}{N_p} = \frac{2}{I_s}$ $I_s = 1A$
35 36	A circular coil of 50 turns has radius 20cm, carries a current 3A. The plane of the coil makes angle 60° to magnetic field whose flux density uniform of 0.5 T. Calculate the magnetic torque acting on the coil	$\tau = B I A N \sin \theta$ $\tau = 2 \times 3 \times \pi (0.2)^2 \times 50 \times \sin (30)$ $\tau = 18.84 \text{ N.m.}$
37 38	A square loop of 10 turns is placed in a uniform magnetic field perpendicular to the plane of the loop. If the magnetic flux changes from 0.03 Wb to zero in 0.1 s. Calculate the average induced EMF in the loop	$\text{emf} = \frac{-N \Delta \Phi_m}{\Delta t}$ $\text{emf} = \frac{(-10)(0-0.03)}{0.1}$ $\text{emf} = 3V$
39 40	When the current changes from + 2A to - 2A in 0.05 second, an e m f of 8 V is induced in a coil. Calculate the coefficient of self-induction in the coil.	$\text{emf} = -L \times \frac{\Delta I}{\Delta t}$ $8 = -L \times \frac{(-2 - 2)}{0.05}$ $L = 0.1 \text{ H}$
41 42	A hot body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å. Calculate the temperature of the sun.	$\lambda_{\max} \propto \frac{1}{T}$ $\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$ $\frac{20000}{5000} = \frac{T_2}{1500}$ $T_{\text{sun}} = 6000 \text{ K}$

باقي الإجابة في الورقة التالية

Group 6:

Compare between each of the following

43 44	Point of comparison	The electric generator	The electric motor
	The scientific base	Electromagnetic induction	Magnetic torque acting on a coil carrying current placed in a magnetic field

45 46	Point of comparison	Cathode Ray tube	The electric transformer
	The use	used in TV and computer monitors	Converting low alternating voltage into high alternating voltage and vice versa. Or It transfers electric power from one circuit to another

47 48	Point of comparison	Direct current (DC).	Alternating current (AC)
	The direction	One direction	Two opposite directions

49 50	Point of comparison	Magnetic dipole moment	Magnetic permeability
	Measuring unit	$A.m^2$ OR $N.m/T$	$T.m/A$ OR $Wb/A.m$

انتهت الإجابة